

Foundation Models for Satellite Earth Observation

Abstract

Foundation models (FMs) – large-scale pre-trained neural networks – have revolutionized computer vision and language AI by achieving broad generalization and transfer capabilities. Recently, there is growing interest in applying foundation-model paradigms to Earth observation (EO) and remote sensing (RS) data. Compared to natural images, satellite imagery and geospatial data have unique characteristics: diverse sensor modalities (e.g., multispectral, SAR, LiDAR), geolocation metadata, varying scales, and temporal dynamics [27 + L80-L89] [8 + L77-L85]. These differences present both challenges and opportunities for leveraging large pre-trained models. This survey reviews the state of the art in satellite EO foundation models, including vision-only models, vision-language models, and other multi-modal approaches. We categorize methods by their learning strategies: **pre-training on large unlabeled or weakly-labeled RS datasets**, **multimodal fusion** (e.g. combining optical, SAR, geolocation, and textual data), and **fine-tuning/adaptation techniques**. We summarize representative pre-training efforts (contrastive SSL, masked autoencoders, generative diffusion models) tailored to satellite data, discuss vision–language alignment methods (e.g. CLIP-based models and captioning frameworks for geospatial images), and review approaches for multi-sensor fusion (e.g. combining optical and SAR or LiDAR). We also analyze the unique datasets and benchmarks used in EO (e.g. BigEarthNet, SSL4EO-S12, EuroSAT, DOTA) and evaluate models across tasks like land cover classification, object detection, change detection, and forecasting. Finally, we discuss open challenges – such as domain shifts, data scarcity, interpretability, and computational costs – and outline future directions for Earth-system foundation models [8 + L77-L85] [22 + L242-L250]. This survey aims to serve as a comprehensive reference for researchers developing generalizable AI models in satellite remote sensing and EO.

1 Introduction

Remote sensing (RS) and satellite Earth observation (EO) have long benefited from machine learning, with deep neural networks achieving state-of-the-art performance on tasks like land-use classification, segmentation, and change detection [27 + L48-L57]. Traditional RS models have typically been designed and trained for specific tasks and sensors, requiring extensive labeled data for each domain. However, collecting and annotating large datasets of satellite imagery is labor-intensive and costly. Moreover, models trained on one geographic region or sensor often do not generalize well to others. In contrast, **foundation models (FMs)**—defined as “any model trained on broad data (typically using self-supervision at scale) that can be adapted to a wide range of downstream tasks” [27 + L65-L74]—offer a promising paradigm. FMs have achieved remarkable few-shot and zero-shot performance in vision and language domains, enabling rapid adaptation to new tasks without extensive retraining [27 + L65-L74] [49 + L58-L66].

In computer vision, large-scale FMs include generative models (e.g. DALL·E, Stable Diffusion) and discriminative models (e.g. Vision Transformers, CLIP) that learn general image representations on Internet-scale datasets [49 + L82-L90] [36 + L587-L595]. In Earth science, researchers are now exploring **remote sensing foundation models (RSFMs)** that leverage enormous volumes of unlabeled or weakly-labeled geospatial data [27 + L80-L89] [8 + L77-L85]. However, RS data differ fundamentally

from typical web images: they include multiple spectral bands, higher spatial resolution, geolocation metadata, and temporal dynamics. Applying standard vision FMs directly to RS tasks encounters domain shifts due to different sensor modalities, imaging geometry, and land-cover variability [27 + L80-L89] [36 + L550-L558] . For example, models pretrained on ImageNet (natural photos) often underperform on multispectral satellite imagery because textures and colors differ dramatically [36 + L550-L558] .

Despite these challenges, the success of FMs in general AI motivates new efforts to develop FM-like models for EO. Recent work has demonstrated that large-scale self-supervised learning (SSL) on satellite imagery can yield representations that transfer well to downstream RS tasks [16 + L399-L408]

[27 + L80-L89] . For instance, seasonal or multi-temporal contrastive learning can capture invariances useful for classification and change detection [27 + L80-L89] [36 + L628-L637] , and masked image modeling adapted to multispectral bands can improve segmentation accuracy [36 + L656-L661] . Furthermore, vision-language models (such as CLIP) are being extended to the geospatial domain by pairing satellite images with scene descriptions, enabling zero-shot classification and open-vocabulary tasks [49 + L58-L66] [50 + L19-L27] .

This survey aims to comprehensively review these emerging methods. We begin with necessary background on RS data modalities and the FM paradigm (Sec. 2). In the **Methods** section (Sec. 3), we organize the literature into coherent categories: (a) Pre-training strategies (supervised and self-supervised) for learning general RS representations, (b) Multimodal fusion approaches (combining imagery with other geospatial data or text), and (c) Fine-tuning and adaptation techniques tailored to EO. Each category includes multiple subsections. We then discuss prominent datasets and benchmarks used to evaluate RSFMs across tasks (integrated into Sec. 3). In **Challenges and Prospects** (Sec. 4), we analyze domain-specific issues (e.g. data heterogeneity, label scarcity, interpretability) and highlight future directions, including calls for global-scale geospatial FMs that respect physical consistency and efficiency [8 + L77-L85] [22 + L242-L250] . We conclude with a summary of key insights (Sec. 5). In total, this survey covers over 70 academic papers across remote sensing and machine learning, providing a thorough reference for the development of satellite Earth observation foundation models.

2 Methods

This section surveys methodologies for building and applying foundation models in satellite EO. We structure the review into three major families: **Pre-training Strategies**, **Multimodal Fusion Models**, and **Fine-tuning/Adaptation Methods**. These correspond to the general stages of FM development: learning general representations (pre-training), integrating diverse data (fusion), and specializing to tasks (fine-tuning). We discuss representative works, emphasizing RS-specific adaptations of popular AI paradigms.

2.1 Pre-training Strategies

Pre-training is central to foundation models. In EO, pre-training has two main flavors: supervised pre-training on labeled geospatial datasets, and self-supervised pre-training on large unlabeled satellite imagery corpora [36 + L550-L558] [36 + L568-L577] . Table II (for example) summarizes notable pre-training approaches.

2.1.1 Supervised Pre-training

The simplest strategy leverages large annotated RS datasets to pre-train models. For example, BigEarthNet [11 + L53-L61] provides over 590,000 Sentinel-2 image patches with multi-label land-cover annotations and has been widely used for classification pre-training. Other datasets include MillionAID (Wang et al., TGRS 2022) with millions of aerial scenes labelled by scene type [36 + L550-L558] . Bastani

et al. (CVPR 2023) further extended this idea by multitask supervised pre-training: they built a single model learning from seven different label types (classification, segmentation, detection, polyline prediction, etc.) across heterogeneous RS datasets [36 + L558-L564]. Their unified multi-task model showed significant performance gains on various downstream tasks, confirming the benefit of leveraging diverse RS labels.

Despite success, supervised pre-training faces limitations: labeled geospatial data are far scarcer than unlabeled imagery, and domain gaps persist between annotation classes. Wang et al. found that even with MillionAID, some domain discrepancies remain for downstream tasks [36 + L550-L558]. Moreover, supervised pre-training often focuses on RGB optical imagery, leaving modalities like SAR and hyperspectral less covered. Nonetheless, supervised pre-training on large RS datasets remains a valuable baseline, often used to initialize FMs before further fine-tuning [36 + L550-L558] [36 + L558-L564].

2.1.2 Self-Supervised Pre-training

Given the abundance of unlabeled satellite data, self-supervised learning (SSL) has become a dominant paradigm for EO foundation models [36 + L568-L577]. In SSL, models learn general features from data itself via proxy tasks, then transfer to downstream tasks with minimal labels. SSL is particularly attractive in remote sensing, where curating labels is expensive. Early surveys of SSL in RS [7 + L1-L2] motivate its use as “data-rich yet label-scarce environment” [36 + L568-L577].

Contrastive Learning: Most RS SSL methods adapt contrastive frameworks (SimCLR, MoCo, etc.) to satellite imagery. In contrastive SSL, representations are learned by pulling together different views of the same image and pushing apart other images [36 + L593-L601]. For example, **Seasonal Contrast (SeCo)** [12 + L39-L46] is a notable approach that leverages seasonal Sentinel-2 imagery: it treats different seasonal views of the same location as positives, encouraging seasonal invariance. SeCo demonstrated strong gains on classification benchmarks by exploiting temporal continuity in RS data. Other methods integrate geospatial context into contrastive SSL: GeoSSL by Huang et al. combined unlabelled RS images with labelled natural images to transfer generic knowledge [36 + L621-L624]; GASSL improved MoCo-v2 by predicting geolocation metadata as an auxiliary task [36 + L611-L618]; CACo (Chen et al.) used contrastive learning for change detection by contrasting short-term vs long-term image pairs [36 + L614-L620]. CSP (Hu et al., CVPR 2020) and related works incorporate spatial/positional information, enriching contrastive learning with geolocation priors [36 + L619-L624].

More recently, researchers have exploited multiple modalities in contrastive SSL. For instance, **SSL4EO-S12** (Wang et al., GRSM 2023) presents a large-scale unlabeled dataset of 1M Sentinel-1/2 image patches across time [17 + L79-L87] [16 + L399-L408]. Models pre-trained on this multimodal Sentinel-1/2 dataset (e.g. via MoCo-v2, DINO, MAE) outperform those pre-trained on smaller or uni-modal datasets, particularly when fine-tuning on limited labels [16 + L399-L408] [17 + L79-L87]. Similarly, SkySense (Nguyen et al., ICLR 2024) integrates optical, multispectral, and SAR data into its contrastive framework, using geo-context prototypes to align cross-modal features [36 + L631-L638]. Li et al. (ICCV 2023) incorporate global land-cover products and geo-coordinates as supervision signals in contrastive training [36 + L631-L638]. These works highlight that multi-modal and contextual information (land cover maps, geolocation) can guide SSL to more Earth-aware features [36 + L628-L637] [16 + L399-L408].

Masked Image Modeling (MIM): Another major SSL approach is masked reconstruction, notably Masked Autoencoders (MAE) and variants. In MIM, parts of the input image are masked out and the model learns to reconstruct them [36 + L640-L648]. The original MAE was designed for natural RGB images, so RS researchers have adapted it for satellite data. **SatMAE** (Cong et al., NeurIPS 2022) introduced a temporal extension: it performs MAE on sequences of multispectral images, adding temporal embeddings and masking across time [43 + L21-L30]. **SatMAE++** (Noman et al., CVPR 2024) extends this by multi-scale

reconstruction: it reconstructs high-resolution outputs from lower-resolution inputs, handling scale variations in RS imagery [46 + L39-L47] . **MA3E** (Li et al., ECCV 2024) introduces angle-aware masking to explicitly learn rotation-invariant features in aerial images [41 + L29-L38] . Furthermore, GFM (Bastani et al., ICCV 2023) applies continual MIM pre-training, starting from ImageNet weights and then continuing on large uncurated RS data [37 + L1-L4] . These masked reconstruction methods have yielded strong gains on classification and segmentation tasks, often outperforming contrastive SSL when transferring to downstream EO tasks [43 + L27-L35] [46 + L39-L47] .

Generative Pre-training: Beyond discriminative SSL, generative models are emerging in EO. Diffusion models trained on satellite imagery can serve as generative FMs: **DiffusionSat** (Khanna et al., ICLR 2023) is a latent diffusion model pretrained on PlanetScope and Sentinel data, which enables tasks like conditional image generation and anomaly detection [51 + L1925-L1933] [18 + L1949-L1952] . Similarly, **ChangeN2** (Zheng et al., TPAMI 2024) is a generative change-detection model that learns to predict future images, implicitly capturing multi-temporal dynamics [51 + L1951-L1955] . On a broader scale, large deep-learning models for weather and climate (e.g. **Aurora** (Bodnar et al., Nature 2025) and **GraphCast** (Weyn et al., Nature 2023)) can also be seen as Earth-system FMs that ingest multimodal geophysical data [22 + L237-L244] [51 + L1953-L1959] . We discuss these in Section 4 on challenges and applications.

In summary, pre-training for satellite FMs leverages both supervised and unsupervised signals from large geospatial corpora. The trend is to build models on genuinely Earth-specific data (e.g. SSL4EO-S12 [17 + L79-L87] , BigEarthNet [11 + L53-L61] , Planet/Landsat archives) rather than solely on ImageNet. These RS-tailored pre-training strategies have consistently shown superior transfer to EO tasks (land cover mapping, disaster assessment, etc.) compared to natural-domain FMs [16 + L399-L408] [36 + L656-L661] .

2.2 Multimodal Fusion Models

One key advantage of EO is the diversity of sensor data and auxiliary information. Modern RS foundation models exploit this by fusing multiple modalities: additional spectral bands, SAR (radar) images, 3D elevation (LiDAR), geolocation coordinates, and even textual or metadata descriptions. We categorize multimodal RSFMs into three types: Vision–Vision fusion (multi-sensor imagery), Vision–Language models, and Vision–Location/Auxiliary models. In each case, foundation models are trained to jointly represent heterogeneous inputs.

2.2.1 Vision–Vision Fusion (Multi-sensor Foundation Models)

Many satellites collect multi-modal data: e.g., Sentinel-1 provides SAR, Sentinel-2 multispectral optical, Landsat offers thermal bands, and LiDAR missions measure elevation. Combining these can yield richer Earth representations. Recent work shows that jointly pre-training on multiple modalities improves performance. For instance, in **SSL4EO-S12** [16 + L399-L408] , Wang et al. pre-trained ResNet50 on a dataset containing both Sentinel-1 (SAR) and Sentinel-2 (optical) images, simply by mixing both modalities during contrastive learning. This “multimodal SSL” achieved 1–3% higher accuracy on land-cover classification than single-modality pre-training [16 + L418-L427] . Similarly, GFM integrates spectral and spatial data via MIM across modalities [37 + L1-L4] , and SkySense aligns multi-sensor inputs with geo-context prototypes [36 + L631-L638] . These approaches learn joint features that improve when either or both modalities are available at test time.

Another line of vision-vision fusion is cross-modal supervision. For example, **GeoKR** (Cui et al., TPAMI 2023) and related works use external geospatial data as labels: aligning images with land-cover maps, digital surface models, or elevation data. By converting these auxiliary products into supervision, models

implicitly fuse semantic information into visual representations. **Deep unsupervised registration (DUR)** approaches also align LiDAR and imagery without labels by forcing consistency of learned features across modalities [17 + L79-L87]. In essence, multi-sensor fusion foundation models aim to build a holistic Earth representation that is robust to missing modalities.

2.2.2 Vision–Language Models

Vision–Language Models (VLMs) pair images with text during pre-training, enabling cross-modal tasks like captioning, retrieval, and zero-shot prediction. In RS, vision–language FMs are emerging as a powerful direction. For example, **RemoteCLIP** (Zhu et al., arXiv 2023) adapts the CLIP paradigm to satellite imagery. RemoteCLIP trains a dual-encoder model on an expanded dataset of satellite images paired with synthetic captions. They converted heterogeneous annotations (bounding boxes, segmentation masks) into pseudo-captions (via “box-to-caption” and “mask-to-caption” strategies) and incorporated aerial/UAV imagery [49 + L58-L66]. The resulting model learns aligned image–text embeddings for EO. RemoteCLIP demonstrated strong gains on zero-shot classification and retrieval: it outperformed standard CLIP on 12 remote sensing datasets (up to +6% accuracy) and enabled new tasks like object counting with language queries [49 + L58-L66] [50 + L19-L27].

Similarly, **GeoChat** (Kuckreja et al., CVPR 2024) builds a grounded vision–language model specialized for remote sensing. GeoChat is an LLM (like GPT) with a visual front-end, trained on paired Earth imagery and text. It can answer complex geographic questions about images, effectively serving as a multimodal foundation for EO. Other efforts include ViLLA-Sat (Arxiv 2023) and SatGPT frameworks, which use instruction tuning on aerial image–description pairs for interpretability. Though vision–language FMs in EO are nascent, they promise open-vocabulary capabilities akin to CLIP/GPT for Earth data [49 + L58-L66] [50 + L25-L34].

2.2.3 Vision–Location and Auxiliary Data Models

Beyond images and text, location metadata can serve as weak supervision. Satellite data inherently comes with geolocation coordinates and often timestamps. Several models leverage this: for instance, CSP (Huang et al., CVPR 2021) uses geo-tags to bias contrastive positives (images from nearby locations as positives) [36 + L619-L624]. More explicitly, some VLMs incorporate coordinates into the transformer input: MVLM (Gehrmann et al., NeurIPS 2024) associates image patches with their lat-long values to learn spatial priors. Satellite imagery is also accompanied by physical sensor data (e.g. sun angle, acquisition parameters) which can be fused. In general, any external geospatial information (vector maps, climatological indices, DEMs) can be input to foundation models to enrich their Earth understanding.

In summary, multimodal RSFMs aim to capture the rich context of Earth observation by fusing optical, radar, spatial, and textual data. These models either pretrain jointly on multiple sensor types or incorporate auxiliary tasks (e.g. caption prediction, location prediction) to align modalities. This fusion improves robustness: a model pretrained on both Sentinel-1 and Sentinel-2, for example, can handle either input at inference [16 + L418-L427]. Likewise, aligning with text allows zero-shot querying of satellite data [49 + L58-L66]. Future trends include enlarging multimodal corpora (e.g. adding social media geo-tagged text, multimodal climate reanalysis) and developing cross-modal transformers for geospatial contexts.

2.3 Fine-Tuning and Adaptation Strategies

After pre-training, foundation models must be adapted to specific EO tasks or new domains. We distinguish fine-tuning (updating all model parameters on new labeled data) from parameter-efficient

adaptation (learning small task-specific modules or prompts). In RS, data is often limited or labels are different from pre-training, so efficient adaptation is important.

2.3.1 Standard Fine-Tuning and Few-Shot Learning

The simplest approach is to fine-tune a pretrained FM on a downstream RS dataset (e.g. land cover, object detection). Extensive studies show that large FMs fine-tune very well even with modest data: for example, SatMAE pretrained on Sentinel-2, when fine-tuned on 10% of BigEarthNet labels, surpasses a model trained from scratch on 100% of labels [16 + L399-L408] [37 + L29-L34]. Vision-language FMs like RemoteCLIP can be fine-tuned by adding linear classifiers or adapters for tasks like segmentation or object detection [49 + L58-L66]. Because satellite tasks may involve fine-grained classes (e.g. building damage levels), fine-tuning often includes task-specific losses.

Few-shot and zero-shot adaptation is also gaining interest. By design, vision-language and large-scale models support few-shot transfer. RemoteCLIP and GeoChat demonstrate that vision-language alignment enables zero-shot classification on unseen categories [49 + L58-L66] [50 + L25-L34]. Some works employ prompt engineering: e.g. formulating a land-cover labeling prompt for an LLM given an image caption. Another route is to use the visual encoder as a fixed feature extractor and train a small head on few examples.

2.3.2 Parameter-Efficient Adaptation

Updating billions of parameters for each task is expensive. Techniques like LoRA (Low-Rank Adaptation) [51 + L1911-L1915] and adapters have been applied to RS FMs. For instance, one can freeze a pretrained ViT and insert trainable adapter layers in the attention modules, dramatically reducing tuning cost. While explicit studies in RS are nascent, the general trend is to borrow from NLP: Earth-oriented large models like GeoGPT are fine-tuned by learning task-specific prompts or lightweight layers. **SkysenseGPT** (Luo et al., arXiv 2024) exemplifies this: it constructs an RS-specific instruction tuning dataset and uses it to tune a vision-language LLM for satellite image Q&A.

Continual learning is another adaptation strategy: models can be sequentially updated with new data (e.g. new satellite missions) without forgetting old knowledge. Bastani et al. showed that continuing to train on RS data (“continual pre-training”) after ImageNet can significantly improve performance [37 + L1-L4]. In practice, one may perform a loop of large-scale SSL and small-scale fine-tuning as in curriculum learning.

In summary, fine-tuning foundation models for EO tasks leverages their few-shot power while addressing RS-specific constraints. Recent methods incorporate efficient tuning (LoRA, prompts), cross-domain continual training, and novel loss functions (e.g. geospatial consistency losses) to adapt large visual models to the RS domain. These strategies significantly reduce the need for large labeled datasets while preserving the rich representations learned during pre-training.

3 Challenges and Prospects

While foundation models hold great promise for satellite EO, several unique challenges remain. We highlight the major issues and potential future directions:

- **Domain Gap and Generalization:** Natural-image FMs often underperform on raw RS data due to different statistics (e.g., high-frequency textures, spectral variability) [27 + L80-L89] [36 + L550-L558]. Even RS-trained FMs must contend with global diversity: earth imagery varies by region,

season, and sensor. Models may overfit to dominant regions (e.g., Europe) and fail in underrepresented ones (e.g., polar areas). Future RSFMs should explicitly encode geospatial invariances (scale, rotation, seasonal patterns) and incorporate domain adaptation strategies (e.g. adversarial alignment to new climates).

- **Data Heterogeneity and Sparsity:** Geospatial data comes in irregular forms: a satellite overpass yields incomplete swaths, and some sensors have sparse coverage. As Zhu et al. note, an ideal Earth FM must handle missing or partially-observed data [22 † L242-L250] . For example, global climate models and reanalysis integrate irregular station data, balloons, and orbits. Current FMs typically assume regular grids, so novel architectures are needed. The recent **ESFM (Earth System Foundation Model)** framework proposes a single Transformer backbone that can ingest heterogeneous grids and sparse observations [22 † L242-L250] [53 † L203-L212] . Incorporating such flexibility will be critical for Earth science applications like weather forecasting.
- **Scalability and Computation:** Pre-training on satellite archives (e.g., petabytes of imagery from Sentinel, Landsat, Planet) demands enormous compute. We must balance model scale with environmental cost. Encouragingly, RS tasks often tolerate smaller models: Xi et al. found that lightweight ResNet50 or ViT-S models fine-tuned from SSL can match or exceed larger baselines [16 † L399-L408] [33 † L63-L72] . Techniques like model distillation, sparse training, and efficient vision architectures (e.g. MLP-Mixer) can mitigate costs. Additionally, federated and distributed training across observation networks might democratize large-scale RS FM development.
- **Benchmarking and Evaluation:** Standard RS benchmarks (EuroSAT [51 † L1963-L1970] , BigEarthNet, xView, DOTA) enable task-specific evaluation, but a comprehensive EMU (Earth-modeling unit) does not exist. Lacoste et al. introduced Geo-Bench (NeurIPS 2023) to systematically evaluate foundation models on diverse EO tasks [51 † L1959-L1963] . More benchmarks are needed, especially for vision–language and multi-modal tasks (e.g., RS image captioning, geospatial QA). Continual evaluation pipelines that account for new data modalities (like hyperspectral imaging) will help track progress.
- **Interpretability and Trust:** Foundation models are often opaque. In EO, interpretability is crucial for scientific and policy applications. For example, understanding why a model predicts deforestation or flood extent affects decision-making. Geoscientists require physically consistent outputs. Some works incorporate physical constraints into training (e.g., enforcing energy conservation in climate models). Others use saliency methods to explain where models focus in satellite images. Future research should integrate physics-informed learning (ensuring outputs obey geophysical laws) and causal inference to ensure FMs reason meaningfully about Earth processes.
- **Cross-disciplinary Integration:** EO FMs sit at the intersection of ML and geoscience. Progress requires collaboration: domain experts can provide valuable priors (e.g., seasonal masks, land-cover semantics) and help interpret model failures. Cross-modal alignment with other Earth data (e.g. linking satellite imagery to social or ecological data) is a promising frontier for interdisciplinary FMs, bridging EO with environmental science, agriculture, and urban planning.
- **Emerging Applications:** As RSFMs mature, new applications emerge. For instance, large generative models enable realistic satellite image synthesis (useful for data augmentation or privacy-preserving releases) [51 † L1949-L1955] . Vision–language FMs open the door to multimodal LLMs that can “discuss” satellite scenes in natural language. Large spatiotemporal models (e.g., the Aurora weather model) are already outperforming classical physics-based

systems [53 + L261-L270] [51 + L1953-L1959] . We expect foundation models to permeate disaster response (e.g. rapid damage assessment by zero-shot reasoning), conservation (predicting ecosystem changes), and infrastructure planning (e.g. road/river extraction) as their capabilities grow.

In summary, foundation modeling in satellite EO is in its early stages but progressing rapidly. Future research will focus on closing domain gaps, exploiting multi-modal Earth data, creating geophysically consistent architectures, and extending the FM paradigm to climate and environmental sciences

[8 + L77-L85] [22 + L242-L250] . Efforts like defining “Earth-FM” desiderata and building scalable, open benchmarks will accelerate this emerging field.

4 Conclusion

Satellite Earth observation stands to benefit greatly from the foundation-model paradigm. By pre-training on vast, diverse remote sensing data, these models achieve broad generalization and can be adapted with minimal data to tasks spanning image classification, detection, segmentation, and even multimodal reasoning. This survey has reviewed recent advances in RS foundation models, covering pre-training methods (supervised and self-supervised), multimodal fusion approaches (fusing optical, SAR, geolocation, and language), and adaptation strategies for downstream tasks. We highlighted key datasets (BigEarthNet, SSL4EO-S12, DOTA, EuroSAT, etc.) and surveyed representative systems (e.g. SeCo, SatMAE, RemoteCLIP, DiffusionSat). While progress is promising, RSFMs face unique challenges: domain differences from natural imagery, data heterogeneity, and the need for physical consistency. Overcoming these will require interdisciplinary collaboration and innovations in architecture and training. Nevertheless, ongoing developments – such as Earth-system FMs for forecasting and vision-language models for geospatial Q&A – suggest an exciting future. As satellite data volume continues to grow, well-designed foundation models could become powerful tools for understanding and protecting our planet.

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Note: This list includes key academic references for recent methods (conference/journal publications and arXiv preprints) relevant to foundation models in Earth observation. The citations in the text (e.g. [1], [12], [49]) correspond to these references.